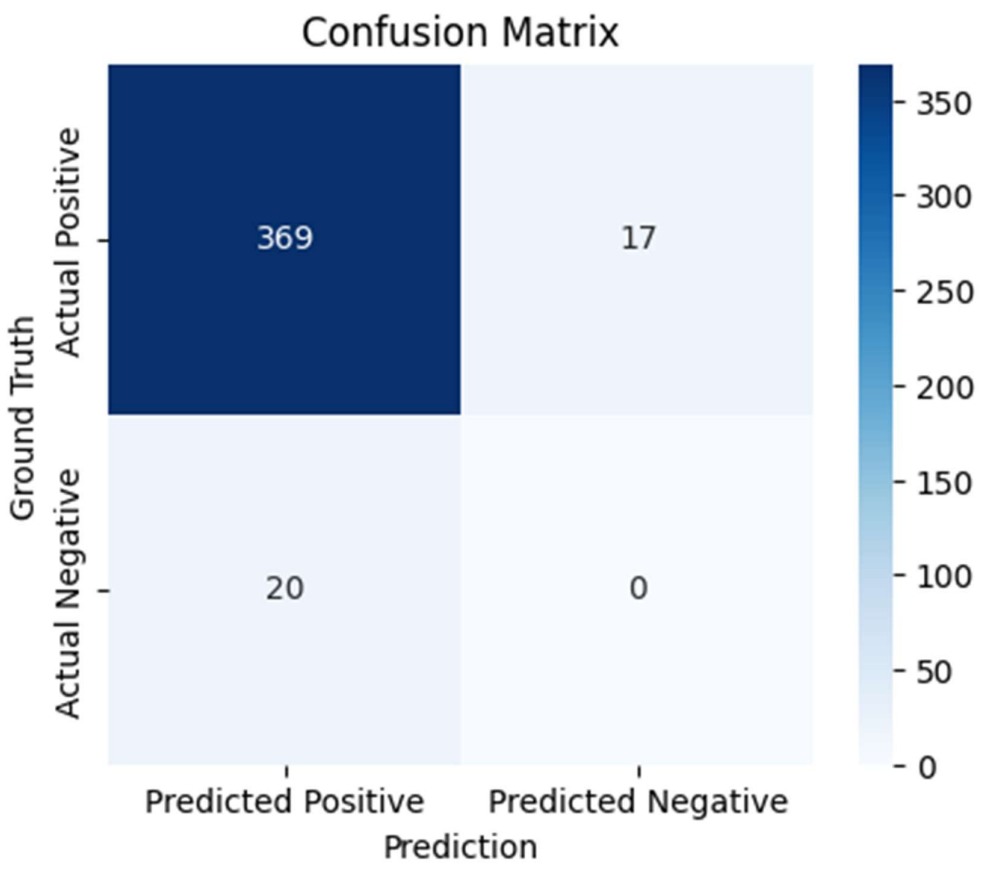
RESULTS

SECTION SEGMENTATION MODEL : MASK R-CNN

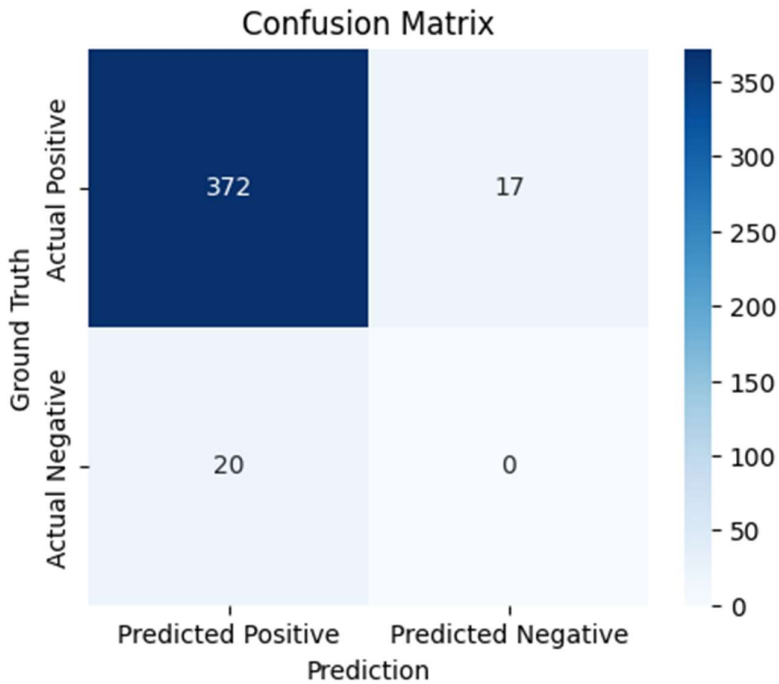
Models	Precision	Recall	F1 Score	mAP50
Bounding Box	**0.9486**	**0.9215**	**0.9523**	**0.9487**
Segmentation Mask	**0.9563**	**0.9489**	**0.9526**	**0.9562**

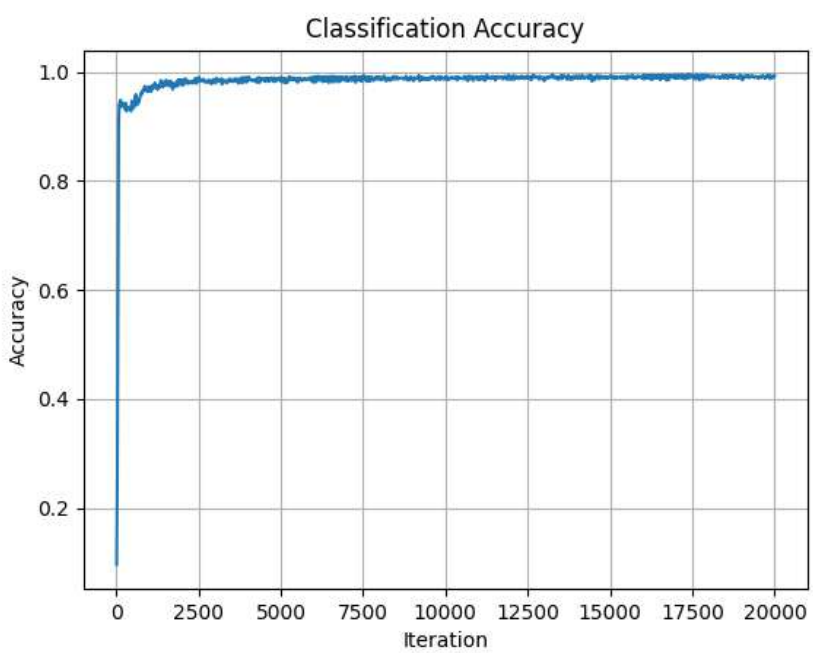
Confusion Matrix :

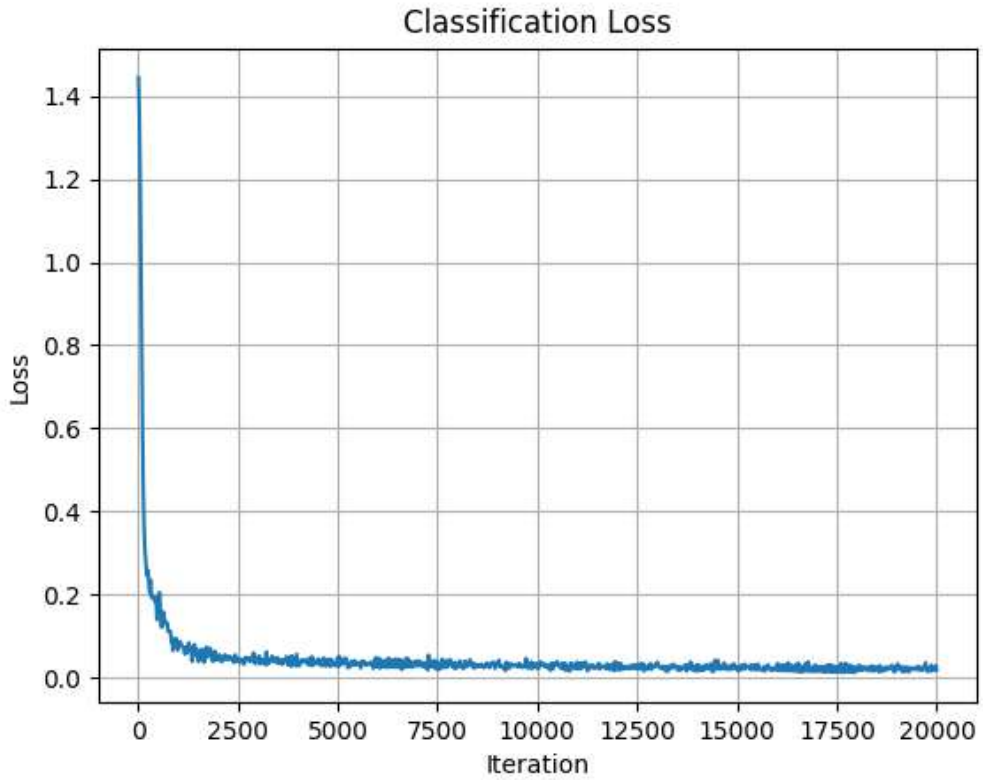
Bounding Box:

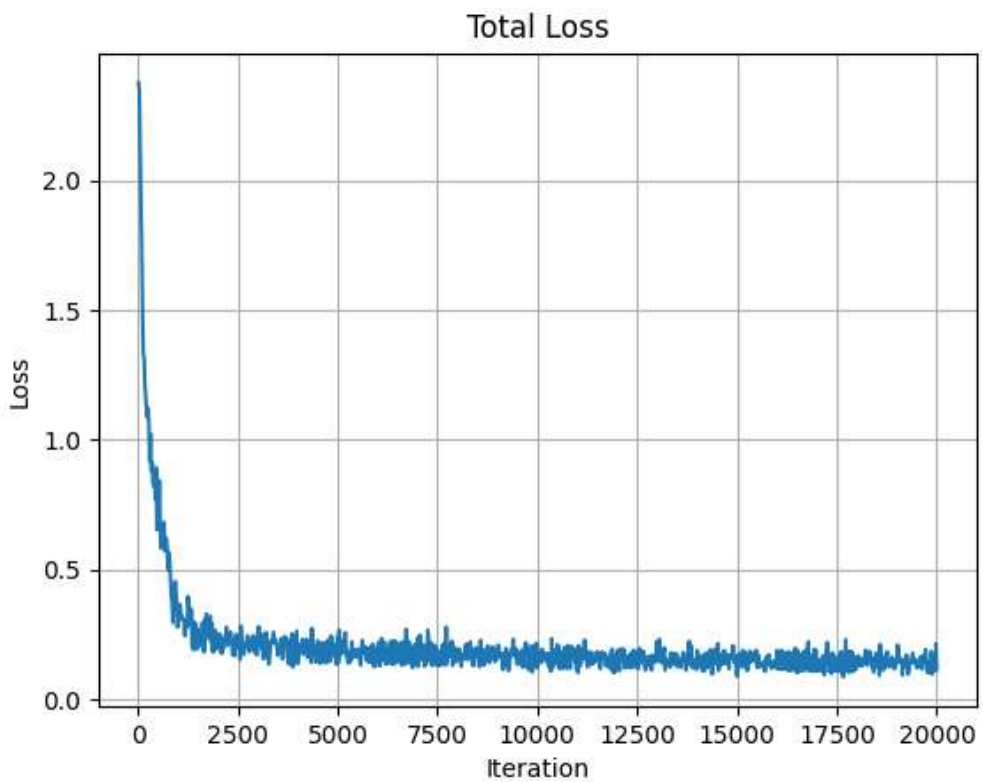


Segmentation Mask:






Classification Loss
Loss
Iteration
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
2500
5000
7500
10000
12500
15000
17500
20000


Total Loss
Loss
Iteration
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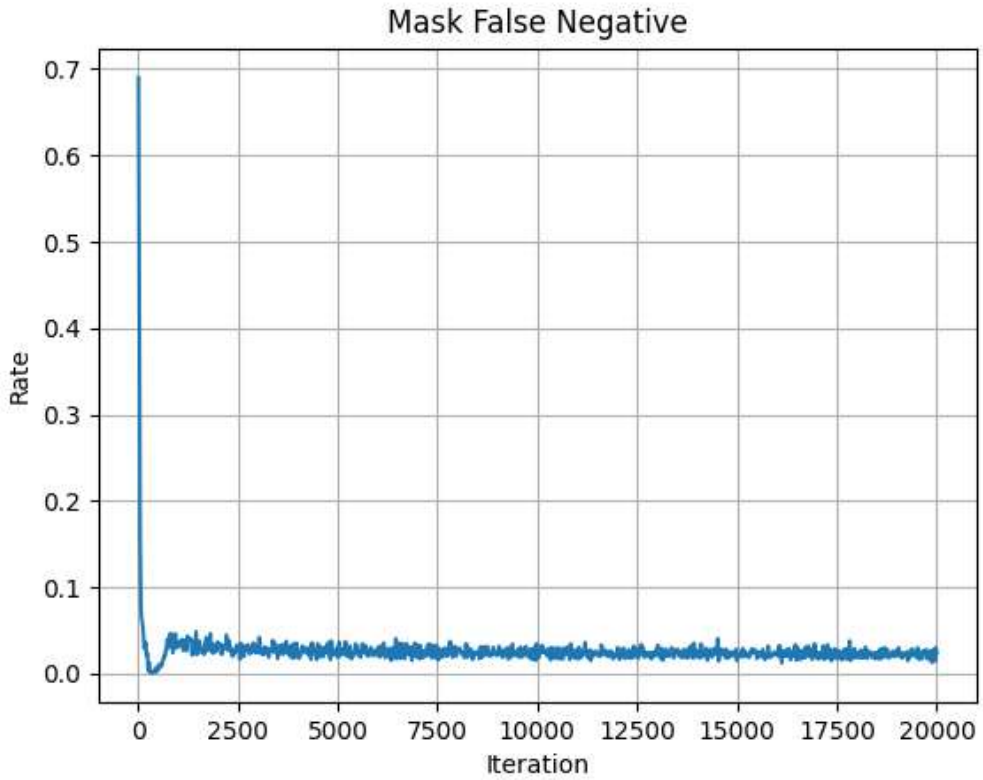
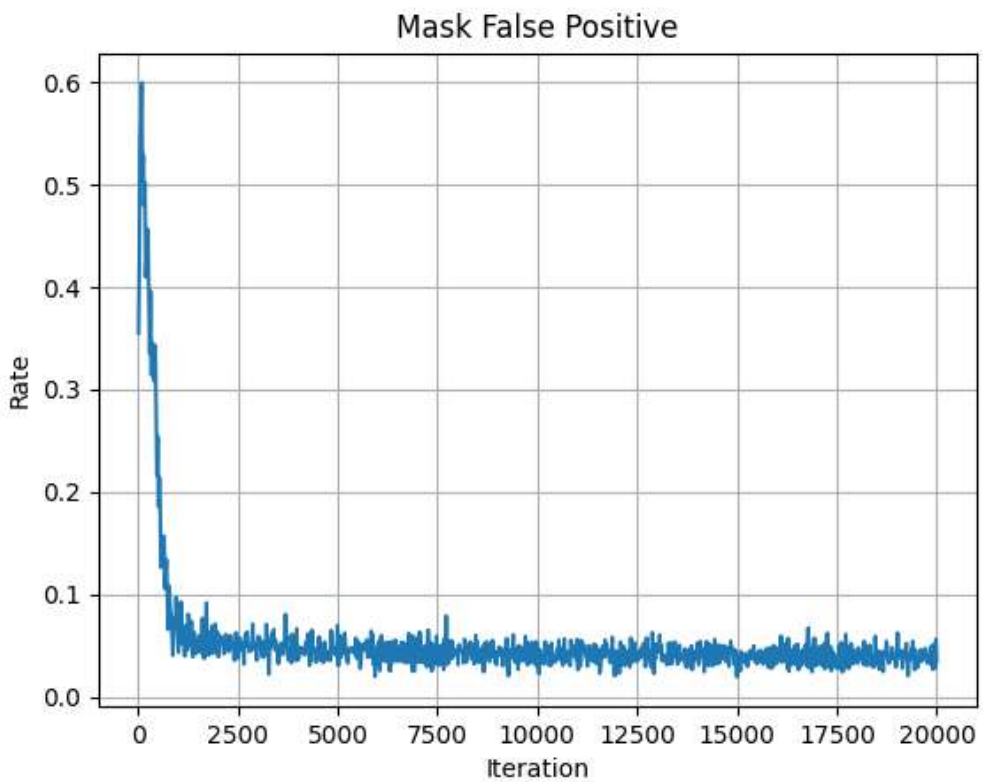
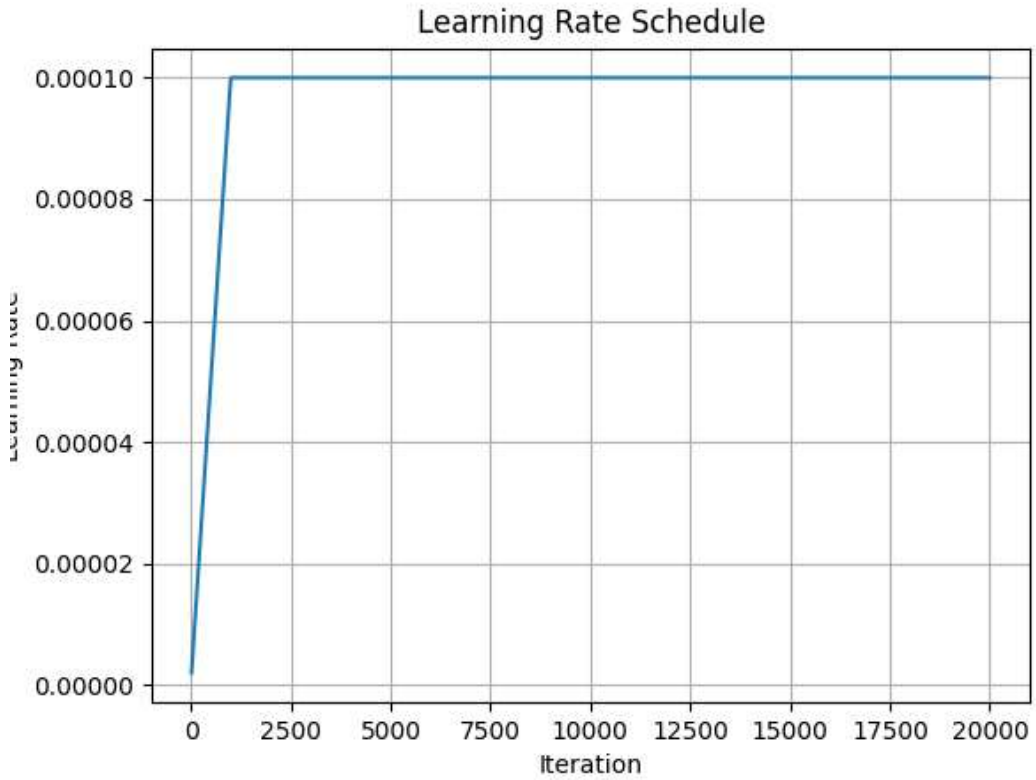
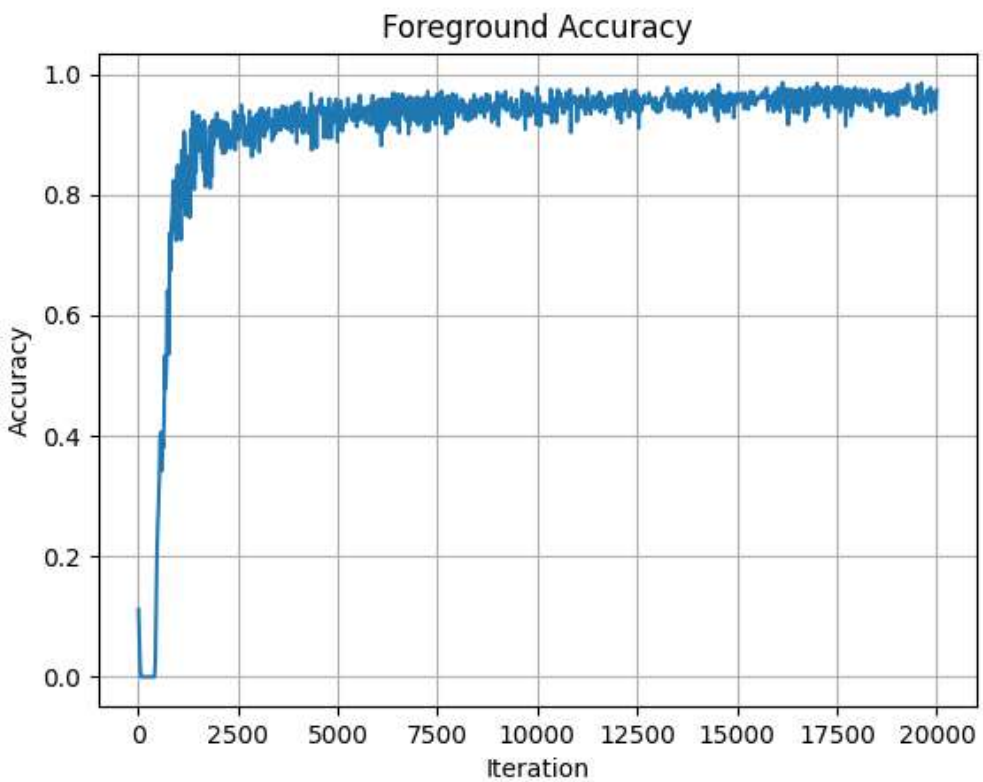
0.5
1.0
1.5
2.0
0
2500
5000
7500
10000
12500
15000
17500
20000


Mask False Negative
Rate
Iteration
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
2500
5000
7500
10000
12500
15000
17500
20000

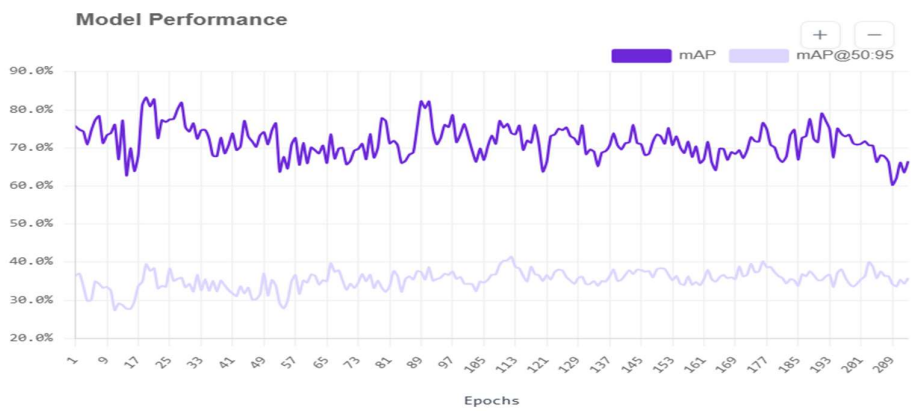

Mask False Positive
Rate
Iteration
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
2500
5000
7500
10000
12500
15000
17500
20000

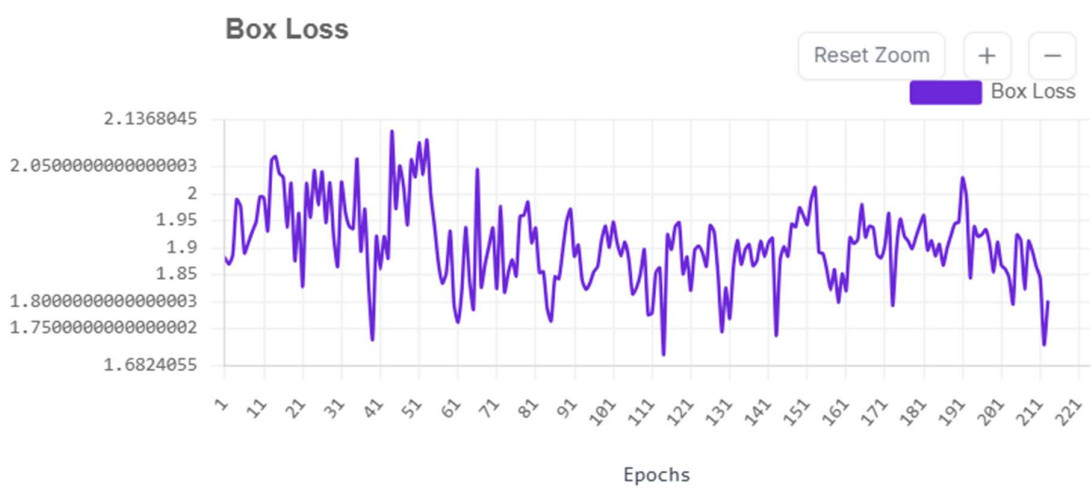

Learning Rate Schedule
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
0
2500
5000
7500
10000
12500
15000
17500
20000
Iteration

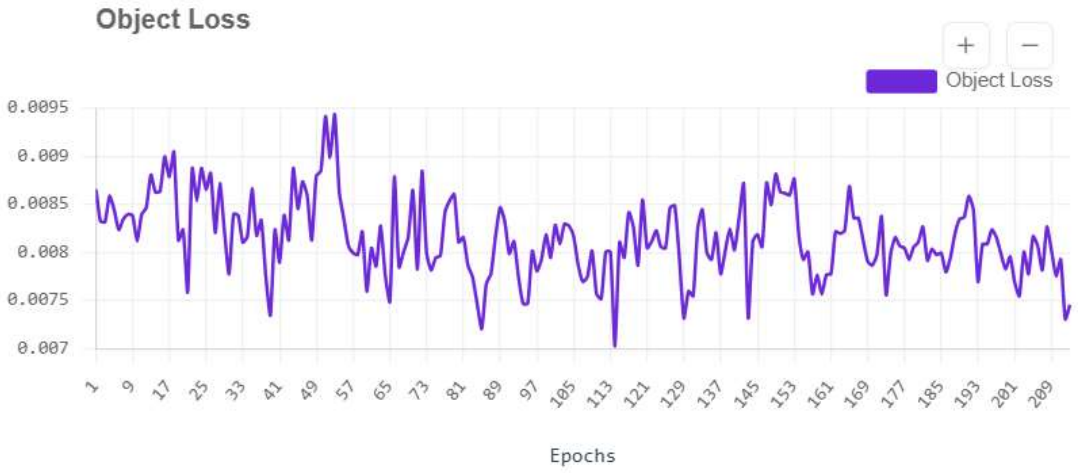

Foreground Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0
2500
5000
7500
10000
12500
15000
17500
20000
Iteration

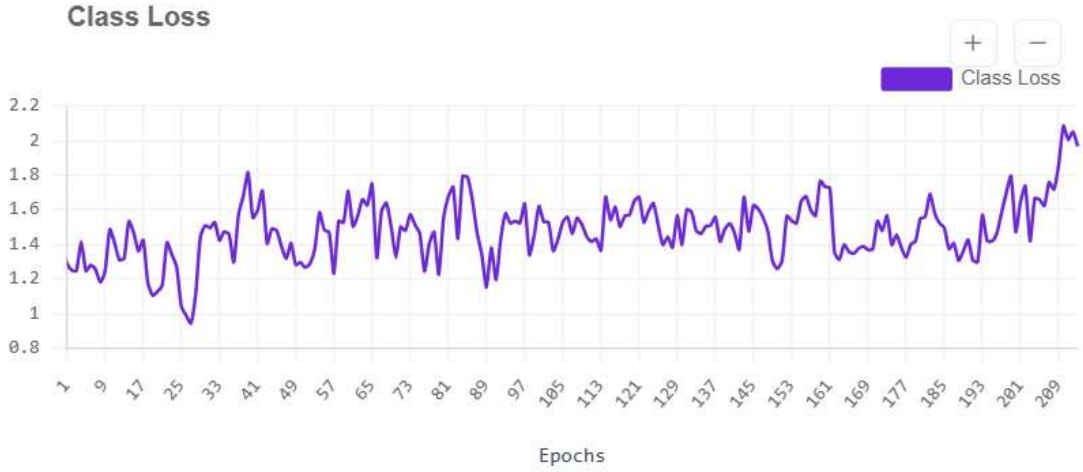
NERVE CANAL SEGMENTATION MODEL :

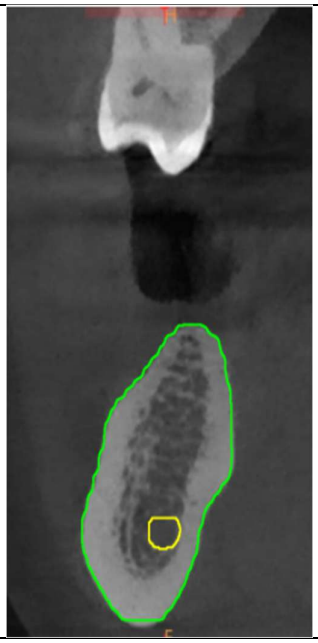
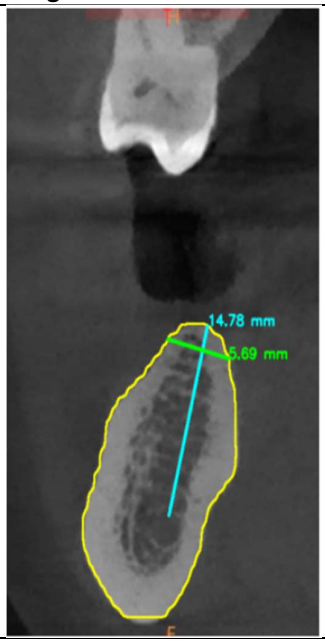
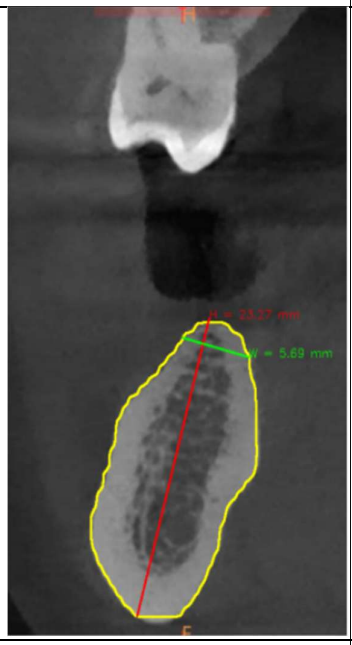
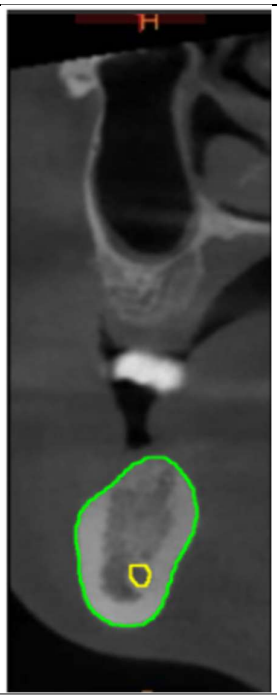
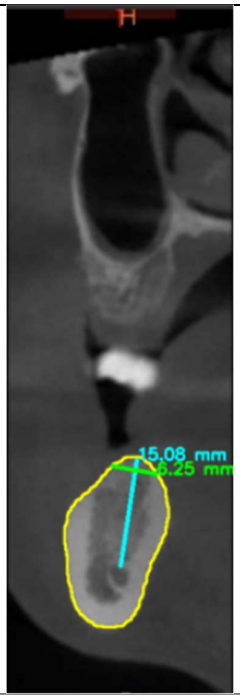
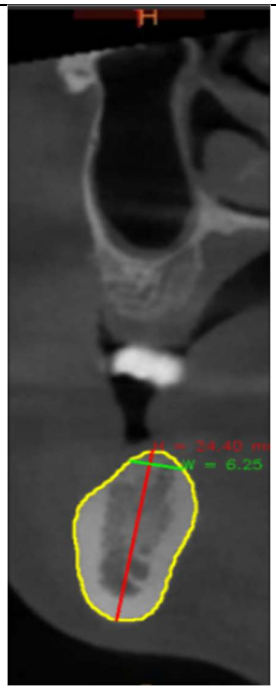
Models	Precision	Recall	F1 Score	mAP@50	mAP@50-95
Yolo26	0.84	0.0621	0.7134	0.748	0.50
Yolov11	0.83	0.682	0.749	0.76	0.54

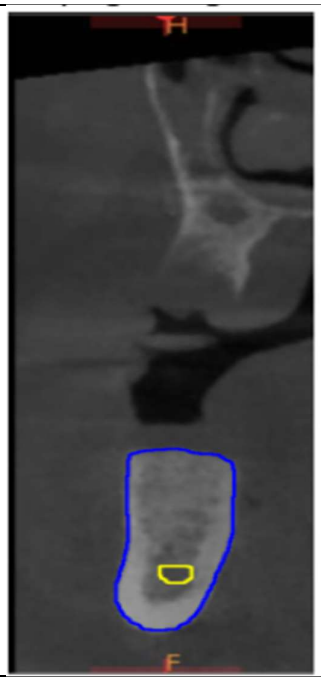
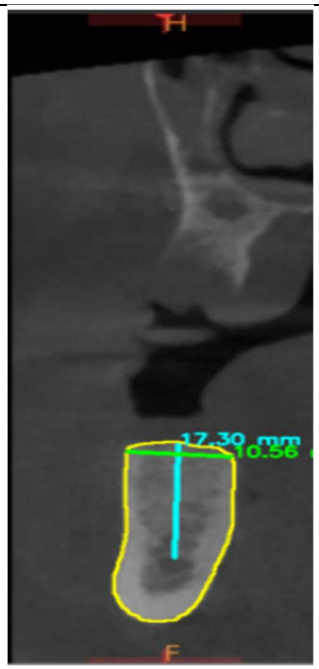
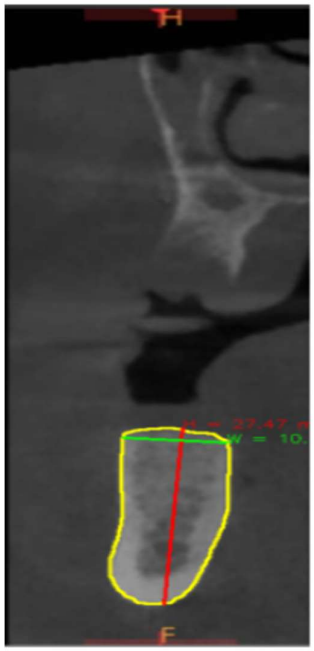
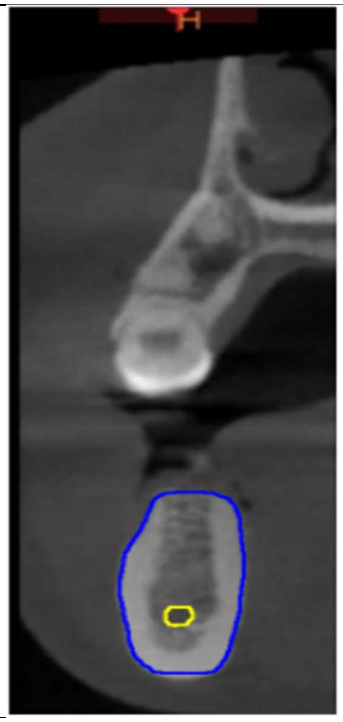
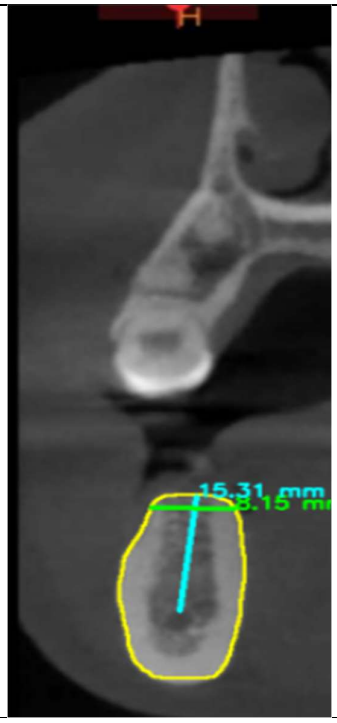
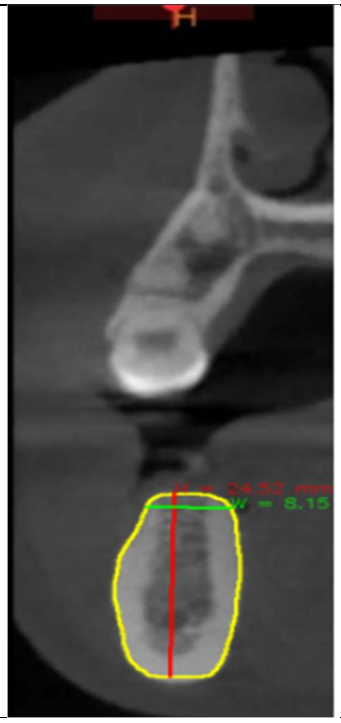





Object Loss
Object Loss
0.0095
0.009
0.0085
0.008
0.0075
0.007
1 9 17 25 33 41 49 57 65 73 81 89 97 105 113 121 129 137 145 153 161 169 177 185 193 201 209
Epochs


Class Loss
Class Loss
2.2
2
1.8
1.6
1.4
1.2
1
0.8
1 9 17 25 33 41 49 57 65 73 81 89 97 105 113 121 129 137 145 153 161 169 177 185 193 201 209
Epochs

	Recommended	**Implant length- 12.78mm**	**Implant Diameter- 2.69mm**	
	Straumann	**Length- 12mm**	**Diameter- 3.5**	
				
	Recommended	**Implant Length- 13.08mm**	**Implant Diameter- 3.25mm**	
	Nobel Biocare **Osstem**	**Length- 13mm**	**Diameter- 3mm**	
				

	Recommended	**Implant Length- 15.30mm**	**Implant Diameter- 7.56mm**
	Nobel Biocare **Osstem**	**Length – 15mm**	**Diameter – 5.5mm**
			
	Recommended	**Implant length- 13.31 mm**	**Implant Diameter- 5.15mm**
	Nobel BioCare **Osstem**	**Length- 13mm**	**Diameter- 5mm**